FROST: Fast Reflex-Like On-Stall Training for Blind Narrow-Stair Climbing in Wheeled-Bipedal Robots

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***Abstract*—Blind narrow-stair climbing requires wheeled-bipedal robots to coordinate stepping when rolling becomes blocked. We present Fast Reflex-Like On-Stall Training (FROST), a task-specific reinforcement learning strategy using event-triggered rewards. A training-time stall-and-posture trigger activates overlapping rewards for body lifting, sequential leg swings and recovery. A supervised history encoder estimates base-frame linear velocity. The deployed policy uses proprioception and its history, commands and previous actions, without terrain sensing, trigger or phase inputs. Under matched training configurations and budgets, FROST reaches higher stair-up curriculum levels earlier than its three ablated variants. Fixed-height simulation tests yield 88% success at 200 mm, compared with 0–2% for the ablations. Prototype demonstrations show traversal of a recessed obstacle and narrow stairs with risers up to 200 mm. These findings support training-time event guidance for learning proprioceptive stair climbing.**

***Index Terms*—Wheeled bipedal robots, blind locomotion, reinforcement learning, asymmetric actor–critic, sim-to-real transfer, stair climbing.**

I. Introduction

Wheeled-legged robots have potential applications in inspection, service, and emergency response because they can roll efficiently while reconfiguring body posture and contacts [1]. Optimization-based methods control legged robots [2]. Wheeled-legged controllers combine rolling, online optimization and gait planning [3], [4]. Learning-based controllers have improved legged and wheeled-legged mobility on irregular terrain [5]–[10]. Continuous ascent on narrow stairs, however, remains difficult for compact wheeled-legged robots. The controller must maintain wheel cruising, detect blockage through body interaction, raise the body, swing both wheel-legs in sequence, and recover balance within a short tread depth.

Blind stair traversal has been demonstrated using cyclic gait-clock inputs on Cassie and an operator-provided stair-mode input on Ascento [11], [12]. Contact-Triggered Blind Climbing (CTBC) uses horizontal contact forces to trigger conditional rewards and adds joint-space feedforward trajectories to the policy actions during training; the feedforward weight is then annealed to zero [13].

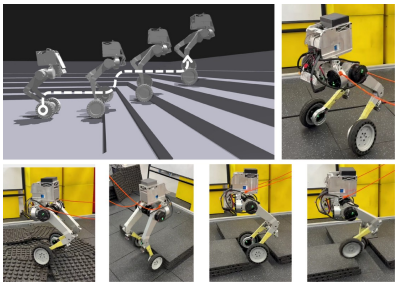


Fig. 1. FROST uses stall-triggered phase guidance during training, then deploys a proprioceptive encoder–actor policy without terrain sensing or phase input.

We propose Fast Reflex-Like On-Stall Training (FROST) for blind continuous narrow-stair climbing on a wheeled-bipedal prototype (Fig. 1). The training-time trigger concentrates stepping rewards on blocked rolling, while overlapping reward windows encourage coordinated body lifting, sequential leg swings and recovery. The design aims to accelerate learning of this transition without adding reference trajectories to policy actions. The deployed policy uses proprioception, commands, previous actions and a history-derived velocity estimate, without trigger, phase, terrain or contact-force inputs. Training builds on privileged-critic actor–critic designs used in prior wheeled-legged stair-climbing work [12], [13].

The prototype has a wheel diameter $D_w = 217$ mm; the tested 200 mm risers reach 92.2% of D_w. Here, "narrow stairs" denotes stairs whose horizontal tread depth d_t, measured along the travel direction, satisfies $d_t/D_w < 1$.

The paper makes two contributions:

1) A task-specific combination of on-stall triggering and overlapping phase rewards, using proximal policy optimization (PPO), asymmetric actor–critic learning and curriculum training. Matched internal ablations assess curriculum progression and fixed-height success rates.
2) Prototype deployment of a proprioceptive encoder–actor policy without trigger or phase inputs, with qualitative demonstrations of recessed-obstacle traversal and stair climbing on the tested robot.

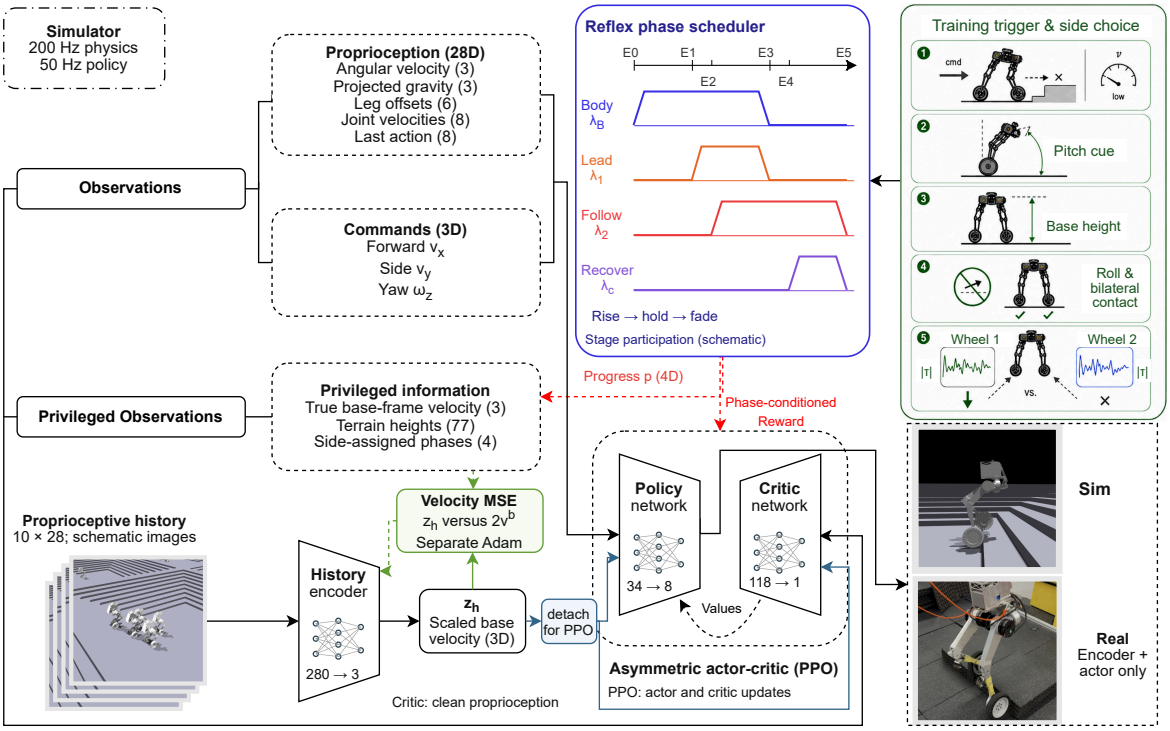


Fig. 2. FROST training and deployment. Trapezoids $\lambda_B, \lambda_1, \lambda_2, \lambda_c$ depict schematic participation of body lift, lead-leg swing, follow-leg swing and recovery in E0–E5 (Fig. 7); they differ from numerical phase progress. The critic input $\boldsymbol{p} = (p_{\text{com}}, p_L, p_R, p_{\text{cool}})$ contains side-assigned progress. $|\tau|$ denotes wheel-torque magnitude. Velocity mean-squared error (MSE) updates the encoder separately, while PPO receives detached estimates. Only the encoder and actor are deployed.

II. Methodology

During training, the stall-and-posture trigger activates phase-guided rewards. Deployment retains only the encoder and actor. Fig. 2 shows the observation partition and learning paths.

A. Robot Prototype and Action Representation

The prototype is a wheeled bipedal robot with two symmetric legs and one wheel at the end of each leg. Each leg has three actuated joints: abduction/adduction, hip pitch, and knee pitch. Together with two wheel joints, the robot has eight controlled degrees of freedom.

At control step t, the dimensionless policy output $\boldsymbol{a}_t \in \mathbb{R}^8$ maps to six leg-position and two wheel-speed targets, following the action split in prior wheeled-bipedal learning [13].

$$\boldsymbol{q}^d_{\text{leg}} = \boldsymbol{q}^0_{\text{leg}} + 0.25\boldsymbol{a}_{\text{leg}}, \qquad \dot{\boldsymbol{\theta}}^d_{\text{wheel}} = 3.0\boldsymbol{a}_{\text{wheel}}. \quad (1)$$

Here, $\boldsymbol{a}_{\text{leg}} \in \mathbb{R}^6$ and $\boldsymbol{a}_{\text{wheel}} \in \mathbb{R}^2$ partition $\boldsymbol{a}_t$; superscript d denotes desired targets (rad and rad/s), and $\boldsymbol{q}^0_{\text{leg}}$ is the default pose. Low-level controllers execute these targets; previous actions remain dimensionless. The measured leg-angle vector is $\boldsymbol{q}_{\text{leg}}$ (also $\boldsymbol{q}_\ell$ in rewards), while $\dot{\boldsymbol{q}}$ contains all eight joint velocities; dots denote time derivatives.

B. Observation and Privileged Information

The actor receives 34 features: 20 sensor-derived values, the previous eight-dimensional action, three commands and a three-dimensional scaled velocity estimate $\boldsymbol{z}_h = E_\phi(H_t)$.

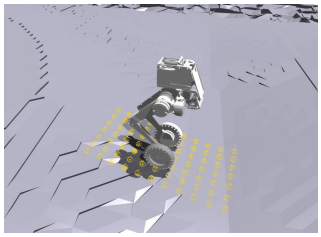


Fig. 3. Critic-only terrain scan. The 77 height samples are used during training and are not provided to the deployed actor.

The history encoder E_ϕ takes H_t, a stack of ten 28-value sensor-and-action frames (280 values). At reset, the current frame is repeated. The critic receives 118 features: the unperturbed 28-value observation, commands (3), the same estimate (3), true base-frame velocity (3), terrain heights (77; Fig. 3) and side-assigned phases (4). Contact forces do not enter either network. Table I specifies scaling and noise.

C. Learning Algorithm

The implementation uses PPO in Isaac Gym [14], [15]. Table II reports the available configuration. Following prior work on supervised state estimation for legged locomotion, the history encoder E_ϕ minimizes the auxiliary velocity loss [16]

$$\mathcal{L}_v(\phi) = \frac{1}{3B}\sum_{i=1}^{B} \left\| E_\phi(H_i) - s_v \boldsymbol{v}^b_i \right\|_2^2, \quad (2)$$

TABLE I
NETWORK INPUTS, SCALING AND ADDITIVE-NOISE HALF-WIDTHS.

Signal	Unit	Dim.	Scale	a_j
Sensor state and previous action (28D)				
Angular velocity $\boldsymbol{\omega}$	rad/s	3	0.25	0.075
Projected gravity $\tilde{\boldsymbol{g}}$	–	3	1	0.075
Leg offsets $\boldsymbol{q}_{\text{leg}} - \boldsymbol{q}^0_{\text{leg}}$	rad	6	1	0.015
Joint velocities $\dot{\boldsymbol{q}}$	rad/s	8	0.05	0.1125
Previous output $\boldsymbol{a}_{t-1}$	–	8	1	0
Additional actor / critic inputs (6D)				
Commands $(v^c_x, v^c_y, \omega^c_z)$	m/s, rad/s	3	S_c	0
Scaled estimate $\boldsymbol{z}_h$	–	3	1	0
Critic-only inputs (84D)				
True velocity $\boldsymbol{v}^b$	m/s	3	2	–
Terrain heights	m	77	5	–
$(p_{\text{com}}, p_L, p_R, p_{\text{cool}})$	–	4	1	–

Units precede scaling; $S_c = (2, 2, 0.25)$ scales forward/lateral speed and yaw-rate commands, respectively. For feature j, a_j is the half-width of independent uniform noise $\mathcal{U}(-a_j, a_j)$, added after scaling and before history stacking. The critic retains the unperturbed 28-value observation. Current actor IMU channels are then replaced by biased-frame values: independent roll/pitch offsets are sampled once per simulated robot from $\mathcal{U}(-1.2^\circ, 1.2^\circ)$. Zero denotes no additional noise, not an error-free estimate.

TABLE II
AVAILABLE NETWORK, OPTIMIZATION AND SIMULATION CONFIGURATION.

Parameter	Setting
Actor / critic hidden layers	$512, 256, 128$ each
Encoder hidden layers	$256, 128$
Environments / rollout steps	8192 / 24 per environment
Epochs / minibatches per epoch	$5/4$, PPO and encoder
Adam initial learning rates	10^{-3}, PPO and encoder
PPO clip / discount / GAE λ	$0.2/0.99/0.95$
Value / entropy weights; PPO gradient norm cap	$1.0/0.01;\ 1.0$
Physics step / action decimation	0.005 s / 4 (50 Hz)
Nominal (K_p, K_d), legs / wheels	$(42, 2.5)/(0, 1)$

Hidden layers use ELU, and the value loss is clipped. PPO adapts its learning rate using KL divergence (target 0.01); the encoder rate is fixed. GAE denotes generalized advantage estimation; λ is its trace-decay parameter. K_p, K_d are proportional and derivative gains, in N m/rad and N m s/rad, respectively.

where i indexes minibatch histories, B is the minibatch size, ϕ denotes encoder parameters, $\boldsymbol{v}^b_i$ is base-frame velocity, and $s_v = 2.0$ empirically scales all three velocity components. A separate Adam optimizer updates the encoder after PPO. Encoder outputs are detached during actor–critic updates and retain the same scaling at deployment.

D. Stall-and-Posture Trigger

At training time, the binary indicator s_t denotes a stall candidate based on command versus base speed M_t, body orientation and height P_t, and bilateral wheel contact C_t. For any Boolean condition A, the indicator function $\mathbb{1}\{A\}$ equals 1 when A is true and 0 otherwise.

$$\begin{aligned} s_t &= \mathbb{1}\{M_t \wedge P_t \wedge C_t\}, \\ M_t &= (|v^c_{x,t}| > 0.3) \wedge (|v^b_{x,t}| < 0.1), \\ P_t &= (|\tilde{g}_{x,t}| > \sin(0.08)) \wedge (z_{b,t} > 0.45) \\ &\quad \wedge (|\tilde{g}_{y,t}| < \sin(0.26)), \\ C_t &= \min_{i\in\{L,R\}} \|\boldsymbol{F}_{i,t}\|_2 > 100. \end{aligned} \tag{3}$$

Here $s_t = 1$ exactly when all conditions in (3) hold. $v^c_{x,t}$ and $v^b_{x,t}$ are commanded and base-frame speeds (m/s); $z_{b,t}$ is world-frame base height (m), and $\boldsymbol{F}_{i,t}$ is simulated wheel-contact force (N). Indices L, R identify left/right wheels. Projected gravity $\tilde{\boldsymbol{g}}$ is expressed in the base frame; its longitudinal/lateral components $\tilde{g}_x, \tilde{g}_y$ constrain pitch/roll.

A wheel-stall timer T^w_t accumulates while $|v^c_x| > 0.3$ and $r_w|(\dot{\theta}_{w,L} + \dot{\theta}_{w,R})/2| < 0.5|v^c_x|$, where $\dot{\theta}_{w,i}$ is wheel i's measured angular speed and $r_w = 0.1085$ m is its radius; otherwise it resets. The combined candidate-or-timeout condition is filtered over five updates:

$$S_t = \prod\nolimits_{k=0}^{4} \mathbb{1}\{(s_{t-k} = 1) \vee (T^w_{t-k} > 1\,\text{s})\}. \tag{4}$$

At each of the five updates, either the complete stall candidate or the wheel-stall timeout can satisfy the filter; the timeout path does not require the posture or contact tests. A trigger starts a cycle only when no cycle is active and resets the wheel-stall timer. The lead leg is selected by the lower sample variance of the last five absolute simulated wheel-torque samples, including the current update; ties select the second wheel in the simulator's joint ordering. This is a contact-stability heuristic.

E. Phase Schedule

Training phase progress. Periodic phase-conditioned rewards have been used in locomotion training [17]. Here, the stall trigger starts a finite 1.1 s training cycle. Let τ_t be the time since the trigger and $m_s(t) \in \{0, 1\}$ indicate whether the cycle is active. The stage set $\mathcal{P} = \{\text{com}, 1, 2, \text{cool}\}$ denotes body lift, first-leg swing, second-leg swing and recovery (cooldown). For each stage k, $t^{\text{start}}_k < t^{\text{end}}_k$ are its start and end times relative to the trigger. Its progress is

$$p_k(t) = m_s(t)\ \text{clip}\left(\frac{\tau_t - t^{\text{start}}_k}{t^{\text{end}}_k - t^{\text{start}}_k}, 0, 1\right), \quad k \in \mathcal{P}. \tag{5}$$

Clipping confines progress to $[0, 1]$: it is zero before a stage starts, rises linearly between its boundaries, and is one afterward while the cycle remains active. At cycle end, the timer and all progress values reset. The lead-leg choice maps the first and second swing stages to the physical legs: $(p_L, p_R) = (p_1, p_2)$ for left-first motion and $(p_L, p_R) = (p_2, p_1)$ for right-first motion. The critic receives $(p_{\text{com}}, p_L, p_R, p_{\text{cool}})$; the actor receives neither phase progress nor the lead-leg flag.

F. Phase-Conditioned Rewards

In a triggered cycle, p_{com} sets the base-lift target and p_i the target for wheel $i \in \{L, R\}$. These targets shape rewards; the deployed actor receives no phases.

TABLE III
REWARD TERMS AND CONFIGURATION COEFFICIENTS BEFORE TIME SCALING AND CLIPPING.

Reward	Formula	Coeff.
Cruise and command-tracking rewards		
Lin. vel tracking	$\exp(-\lVert \boldsymbol{v}^c_{xy} m_{\neg c} - \boldsymbol{v}_{xy} \rVert^2 / 0.25)$	4.0
Ang. vel tracking	$\exp(-(\omega^c_z m_{\neg c} - \omega_z)^2 / 0.25)$	5.0
Orientation	$5\tilde{g}_y^2 + \tilde{g}_x^2$	-2.0
Wrong-direction vel	$[\lvert v_x \rvert - 0.3]_+^2 \mathbb{1}\{O_t\}$	-5.0
Velocity deficit	$\delta_v^2 \mathbb{1}\{\lvert v^c_x \rvert > 0.1\} m_{\neg s}$	-5.0
Rolling-posture and style rewards		
Default pose tracking	$\exp(-\lVert \boldsymbol{q}_\ell - \boldsymbol{q}^{ref}_\ell \rVert^2 / 0.2) m_p$	2.0
Vertical lin. vel	$v_z^2 m_{\neg s}$	-1.0
Action smoothness	$\lVert \boldsymbol{a}_t - 2\boldsymbol{a}_{t-1} + \boldsymbol{a}_{t-2} \rVert^2 (1 - 0.8 m_s)$	-0.05
Leg symmetry	$\exp(-(\lvert y_L \rvert - \lvert y_R \rvert)^2 / 0.005)$	0.5
Excess pitch	$[\tilde{g}_x - 0.12]_+^2$	-10.0
Phase-shaped climbing rewards		
Base-height tracking	$\exp(-[h_b^d - h_b]_+^2 / 0.02) m_s (1 - p_{\mathrm{cool}})$	4.0
Relative swing tracking	$\exp(-(e_h + 2[e_d - 0.05]_+^2) / 0.02)$	3.0
Cooldown standstill	$(\lVert \dot{\boldsymbol{q}} \rVert^2 + \lVert \boldsymbol{v}_{xy} \rVert^2) m_c$	-5.0
Swing wheel brake	$\sum_{i \in \{L,R\}} \mathbb{1}\{p_i > 0.1\} \dot{\theta}^2_{w,i}$	-0.5
Stall clearance	$\mathbb{1}\{0 < \tau_t < 0.8\} \big(20 [\Delta z_b]_0^{0.15} + 2 [v_z^b]_0^{0.8}\big)$	10.0
Step yaw hold	$[\lvert \omega_z \rvert - 0.1]_+ m_s$	-3.0
Soft landing	$\sum_{i \in \{L,R\}} [F_i^z - 300]_+^2 m_s$	-2.0e-4
Sim-to-real regularization rewards		
Torques	$\lVert \boldsymbol{\tau} \rVert^2 (1 - 0.7 m_s)$	-1.0e-4
Joint position limits	$\sum_j ([q_j^- - q_j]_+ + [q_j - q_j^+]_+)$	-5.0
Collision	$\mathbb{1}\{\lVert \boldsymbol{F}_{b_0} \rVert_2 > 30 + 50 m_s\} + \mathbb{1}\{\lVert \boldsymbol{F}_{b_1} \rVert_2 > 5\}$	-2.0
Feet distance	$[d_{min} - d_f]_+^2 + [d_f - d_{max}]_+^2$	-5.0

$[x]_0^u = \mathrm{clip}(x, 0, u)$; $\Delta z_b = z_{b,t} - z_{b,t-1}$. Here $\boldsymbol{v}_{xy} = (v_x, v_y)$ and $\boldsymbol{v}^c_{xy} = (v^c_x, v^c_y)$ are base-frame planar velocity and its command; v_x, v_y, v_z are components of $\boldsymbol{v}^b$, and ω_z is base yaw rate. Forces are world-frame quantities; $\boldsymbol{\tau}$ is the joint-torque vector and F_i^z is wheel i's vertical contact force. y_i is the wheel lateral coordinate in the base frame; d_f is world-xy wheel separation, with $(d_{\min}, d_{\max}) = (0.32, 0.38)$ m. The configured reward pose is $\boldsymbol{q}^{ref}_\ell = (0, 0.3, 0.6, 0, -0.3, -0.6)$ rad in left/right abad–hip–knee order. For joint angle q_j, lower/upper soft limits q_j^-, q_j^+ retain 95% of the modeled range about its midpoint. $\boldsymbol{F}_{b_0}, \boldsymbol{F}_{b_1}$ are contact forces on the first two bodies in the expanded penalized-body list; thresholds are in N.

Base-height target. Let h_j^{terr} be terrain-height sample j, $\bar{h} = 77^{-1} \sum_{j=1}^{77} h_j^{\mathrm{terr}}$ their mean, and z_b the world-frame base height (time indices omitted). The base height above this sampled terrain and its target are

$$h_b = z_b - \bar{h}, \qquad h_b^d = h_0 + A_b \sin(\pi p_{\mathrm{com}}). \tag{6}$$

Wheel-lift target. Let $z_{w,i}$ be the world-frame height of wheel body i. Its nonnegative rise relative to the nominal base-wheel geometry, and the corresponding swing target, are

$$\ell_i = [z_{w,i} - z_b + h_0 - r_w]_+, \qquad \ell_i^d = A_\ell \sin(\pi p_i). \tag{7}$$

Here $[x]_+ = \max(x, 0)$, $h_0 = 0.8585$ m is the nominal base height, $r_w = 0.1085$ m is the wheel radius, and $A_b = A_\ell = 0.30$ m are the target amplitudes. Thus h_b is not mass-weighted center-of-mass height, and ℓ_i is not clearance over a step. Each target returns to its starting level at phase progress 0 and 1 and peaks at 0.5.

Swing errors. Individual lift tracking and left–right coordination are measured by

$$\begin{aligned} e_h &= \sum\nolimits_{i \in \{L,R\}} (\ell_i - \ell_i^d)^2, \\ e_d &= \lvert (\ell_L - \ell_R) - (\ell_L^d - \ell_R^d) \rvert. \end{aligned} \tag{8}$$

The swing reward decreases as e_h increases or e_d exceeds 0.05 m; only e_d has a dead zone. Outside a cycle, $p_i = 0$ and $\ell_i^d = 0$, so this still-active term discourages unintended lift.

Reward activation. The filtered trigger S_t in Section II-D starts a 1.1 s cycle. Its active-cycle mask m_s differs from the instantaneous stall candidate s_t: $m_s = 1$ throughout the cycle and zero otherwise. Two masks used in Table III are

$$\begin{aligned} m_c(t) &= m_s(t) \mathbb{1}\{0.8 \le \tau_t < 1.1\}, \\ m_p(t) &= 1 - m_s(t) + p_{\mathrm{cool}}(t). \end{aligned} \tag{9}$$

Thus m_c activates standstill only in the final 0.3 s. The posture weight m_p equals one outside the cycle and zero at its start, then rises with cooldown progress. Because $p_{\mathrm{cool}} = 0$ outside the cycle and lies in $[0, 1]$ inside it, $m_p \in [0, 1]$ without additional clipping. The base-height term uses $m_s(1 - p_{\mathrm{cool}})$; its height-tracking factor penalizes only $h_b < h_b^d$. A complement $m_{\neg x} = 1 - m_x$ turns a term off when mask m_x is on. Wheel braking starts at $p_i > 0.1$ and lasts until cycle reset.

Reward components. Table III groups command tracking, posture, stair motion and regularization terms. Standard tracking, posture, smoothness, spacing and actuator terms follow prior work [13]. The reward pose $\boldsymbol{q}^{ref}_\ell$ differs from the action default pose. In the command block, let $O_t = (\lvert v^c_x \rvert > 0.1) \wedge (v^c_x v_x < 0)$, with speeds in m/s. The indicator $\mathbb{1}\{O_t\}$ flags motion opposite to the command. The speed shortfall is $\delta_v = [0.5 \lvert v^c_x \rvert - v_x \operatorname{sgn}(v^c_x)]_+$: sgn is the sign function, so velocity along the command is compared with half the commanded speed.

Each table row defines a term $r_i(t)$ and a signed coefficient w_i. With a control interval $\Delta t = 0.02$ s and symmetric clipping $C_b(x) = \mathrm{clip}(x, -b, b)$ with bound $b > 0$, the total reward is

$$r_t = C_{100} \left(\sum_i C_5(\Delta t \, w_i r_i(t)) \right). \tag{10}$$

The factor Δt scales each row to a control-step contribution. C_5 limits each weighted contribution to $[-5, 5]$ before summation; C_{100} limits the sum to $[-100, 100]$. Negative coefficients denote penalties, and negative total rewards are retained.

G. Curriculum and Domain Randomization

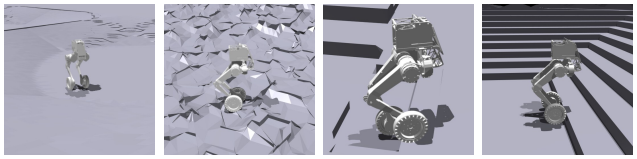
Smooth, rough, obstacle and stair terrains (Fig. 4) are sampled with probabilities $(0.1, 0.3, 0.3, 0.3)$. Difficulty rises with travel and falls after failure, following game-inspired curricula [7]. Table IV lists our configured dynamics-randomization ranges. This approach follows prior sim-to-real control [18]; several ranges also appear in wheeled-bipedal climbing work [13].

H. Deployment Policy

Lead-leg selection influences PPO actor updates through side-specific rewards and the critic's phase inputs. The deployed actor uses proprioception, commands, previous actions and the history-derived velocity estimate, without

TABLE IV
DOMAIN-RANDOMIZATION RANGES USED FOR SIM-TO-REAL TRANSFER.

Parameter	Range or setting	Unit
Friction coefficient	$[0.2,\ 1.6]$	-
Restitution coefficient	$[0.0,\ 0.5]$	-
Base-mass offset	$[-1.0,\ 2.0]$	kg
Center-of-mass shift	$\pm[30,\ 20,\ 30]$	mm
Link inertia scale	$[0.8,\ 1.2]$	-
Push interval	7	s
Max push velocity (xy)	1.5	m/s
Force resampling time	15	s
K_p factor	$[0.8,\ 1.2]$	–
K_d factor	$[0.8,\ 1.2]$	–
Motor torque scale	$[0.8,\ 1.2]$	-
Default joint offset	$[-0.05,\ 0.05]$	rad
IMU roll/pitch offset	$[-1.2,\ 1.2]$	degree
Action delay	$[0,\ 30]$	ms
Joint friction scale	$[0.5,\ 1.5]$	-
Armature scale	$[0.8,\ 1.2]$	-



(a) Smooth slope (b) Rough slope (c) Discrete obstacles (d) Stairs

Fig. 4. Curriculum terrains used to train both wheel cruising and stair-climbing responses.

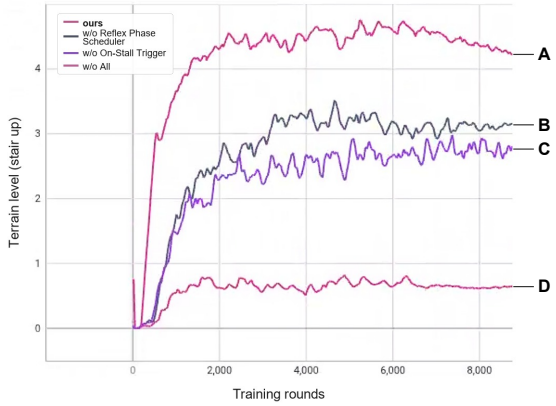


Fig. 5. Stair-up curriculum progression. A: FROST; B: w/o reflex phase scheduler; C: w/o on-stall trigger; D: w/o both components.

trigger, phase or lead-leg inputs. Velocity supervision does not establish explicit phase reconstruction by the encoder.

III. EXPERIMENTS

A. Simulation Ablations

All four variants shared the curriculum, terrain randomization, network architecture and training budget. Without the reflex phase scheduler, a binary stepping signal replaced overlapping body-lift, leg-swing and cooldown rewards. Without the on-stall trigger, activation was periodic; w/o All combined both removals. These comparisons isolate components within FROST rather than benchmark external controllers.

Fig. 5 plots stair-up terrain level against training rounds; each level represents an average 50 mm increase in stair height. FROST reached higher levels earlier and retained the highest final level, despite later fluctuations and decline. Removing both components kept progression near the lowest level. This supports faster curriculum progression, without quantifying wall-clock speedup or variability across training seeds.

TABLE V
SIMULATION SUCCESS RATES (%) AT FIXED STAIR HEIGHTS.

Ablation experiments	Step height (mm)						
	60	90	120	150	180	200	240
FROST (Ours)	**100**	**100**	**100**	**97**	**95**	**88**	**65**
FROST w/o Reflex Phase Scheduler	73	69	45	45	12	2	0
FROST w/o On-stall Trigger	54	52	44	31	4	0	0
FROST w/o All	47	26	3	3	0	0	0

Fixed-height tests (Table V) gave FROST 88% success at 200 mm, versus 2% without the scheduler and 0% for the other ablations. At 240 mm, evaluated only in simulation, FROST achieved 65% versus 0% for all ablations. Thus, the complete design improved both curriculum progression and learned stair-climbing performance within this evaluation.

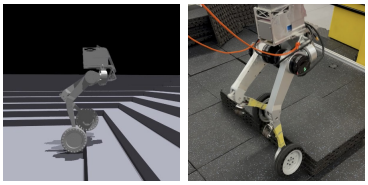
B. Prototype Deployment

Prototype demonstrations examined a 100 mm hole and stairs with 150 mm and 200 mm risers, mainly with approximately 200 mm tread depths. These provide qualitative evidence of deployment feasibility.

Fig. 6 compares body-lifting postures in simulation and on the prototype. Fig. 7 shows two left-first climbing cycles, with snapshots aligned to hip–knee traces. Read with the schematic in Fig. 2, body lift spans E0–E3, lead-leg transfer E1–E3, follow-leg transfer E2–E5, and support recovery E4–E5. The follow-leg transfer therefore overlaps recovery from E4 to E5.

The unchanged policy in Fig. 8 was tested at least 20 times for each riser height and approach: approximately frontal, left-wheel-first and right-wheel-first. Tests began before the stairs; retries within a test were not counted separately. Oblique approaches made one wheel contact earlier. In these tests, frontal and clearly left-wheel-first approaches consistently led with the left leg; clearly right-wheel-first approaches led with the right. Slightly earlier right-wheel contact could still produce left-first lifting.

Forward speed commands were approximately constant. On higher stairs, the robot occasionally lost balance during second-leg transfer; the policy restored balance without phys-



(a) Simulation (b) Real robot

Fig. 6. Representative body-lifting postures in simulation and on the prototype.

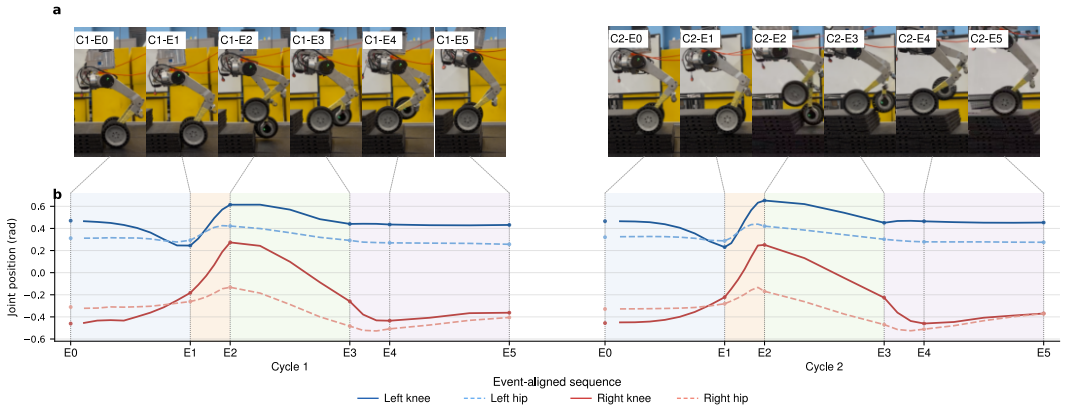


Fig. 7. Two illustrated stair-climbing sequences with hip and knee traces. E0–E5 identify the image events used for piecewise-linear alignment of each trace. Horizontal spacing denotes aligned sequence position rather than elapsed time.

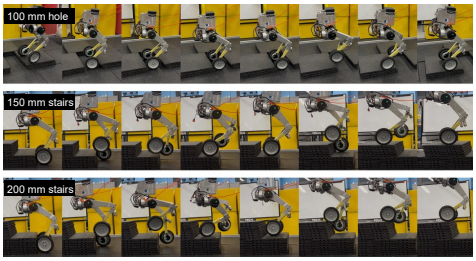


Fig. 8. Prototype traversal of recessed and raised discontinuities using the same proprioceptive policy.

ical support or repositioning. Without closed-loop heading-angle control, recovery could leave a yaw offset requiring operator steering to realign the robot before retrying to complete the ascent.

IV. Discussion and Conclusion

FROST combines stall-triggered rewards with overlapping guidance for coordinated stepping. Internal ablations support earlier curriculum progression and higher fixed-height success rates under matched budgets. Prototype demonstrations support transfer to the tested robot without deployment-time phase inputs.

These findings do not establish superiority over external methods or real-world repeatability. Tests covered one robot and limited stair geometries; recovery could require operator steering before retrying. Sensitivity to phase timing, lift amplitude, friction and payload remains unquantified. Matched controller comparisons and repeated trials are needed to assess broader applicability.

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